

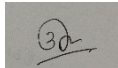


Indian Institute of Management Ranchi

MASTER OF BUSINESS ADMINISTRATION (MBA)
AY 2026-27, TERM -II

TITLE OF THE COURSE: Operations Management -I
CREDITS: 1.5

FACULTY DETAILS

Name (sections teaching)	Faculty Office Address	Email	Signature
Prof. Utkarsh Shivam (2 section)	710, Birsa Munda Block	utkarsh.shivam@iimranchi.ac.in	
Prof. Ranit Sinha (1 section)	703, Birsa Munda Block	ranit.sinha@iimranchi.ac.in	
Prof. Krishna Kumar Dadsena (2 section)	405 - 4th floor, Sidhu Block	krishnakumar.dadsena@iimranchi.ac.in	

CONSULTATION TIME FOR STUDENTS: On Appointment

COURSE DESCRIPTION: -

Operations management creates value in the form of goods and services by transforming inputs into outputs. Operations management is a critical aspect of a business because it involves the efficient and effective management of resources to meet customer demand and achieve organizational goals. The foundation course on Operations management focuses on the understanding of operations management principles and techniques. By the end of the course, students will be equipped with the knowledge and tools to manage operations and drive business success effectively.

COURSE OBJECTIVES: -

Understand the importance of operations strategy aligned with the competitive priorities of an organization that support the decisions needed to manage the firm.

Learning Outcomes and Associated Measures: -

Course Learning Outcome	Traits	Assessment Tool(s)
1.1. Understanding operations and operations management.	(T-9) Students should be able to identify the interconnectedness across business functions	Embedded Question(s) in mid-term, end-term
1.2. Understanding process strategy and process structure in manufacturing and service		

sectors. Selecting a manufacturing/ service process considering strategic fit. 1.3. Understanding operations strategy aligned with competitive priorities.		
2.1. Understand how to configure operations into layouts design. 2.2. Understand the influential factors that affect location decision of a facility. 2.3. Understanding different components of a process and capacity-related factors.	(T-10) Students should be able to propose solutions to problems across different business functions	Embedded Question(s) in mid-term, end-term, Quiz
3.1. Understanding project management, and work breakdown structure. 3.2. Apply Project planning and scheduling	(T-10) Students should be able to propose solutions to problems across different business functions	Embedded Question(s) in mid-term, end-term, Quiz
4. Understanding forecasting techniques according to the data set.	(T-9) Students should be able to identify the interconnectedness across business functions.	Embedded Question(s) in mid-term, end-term and group assignments, Quiz

PRESCRIBED TEXTBOOK:

Operations Management: sustainability and supply chain management, 12th Edition, Pearson. Authors: Jay Heizer, Barry Render, Chuck Munson and Amit Sachan.

EVALUATION: -

This course will be evaluated by three different components. These components are end term exam, group project, and quiz. Each group will be given a separate project. The project provides an opportunity for students to develop decision making skill in operations management.

GRADING SCHEME: -

Component	Mode	Weightage
End Term Exam	Close Book	40%
WAI	Compulsory	25%

Attendance	Compulsory	10%
Quiz	2 Quiz	25%
Total		100%

GROUP ASSIGNMENT DETAILS:

Case 1: *Chinese Pharmaceuticals (HK) Limited: Effective Forecasting for Optimal Inventory Management. The Hong Kong University of Science and Technology. Product #:ST5-PDF-ENG*

Case 2: *Tastee Snax Cookie Company (A). Darden School of Business. Product #:UV5918-PDF-ENG*

SOFTWARE(S) USED: Excel

FIELD TRIPS / WORKSHOPS PLANNED: No

ACADEMIC MISCONDUCT: Please refer to the latest program manual provided to you.

ELEMENTS OF SOCIAL IMPACT IN COURSE CURRICULUM (YES/NO): No

DETAILED SESSION PLAN:

Session No.	Business Problem	Topic(s) & Discussion Question	Readings (Text Book, Articles, Cases)	Pre-Read
1	Organizations face the challenge of managing goods and services efficiently from inception to delivery while improving productivity, resource utilization, and operational performance. Managers must make informed operations decisions to enhance efficiency, quality, and organizational competitiveness.	Topic(s): Operations and Productivity Discussion Question(s): ✓ Objective of Operations Management? ✓ How can operations management help improve business performance? ✓ What is the role of operation managers in a dynamic business environment? ✓ Operations of goods versus services	Text Book: Heizer and Render, Chapter 1, Page: 2-19	On theory in operations management; (Journal of Operation Management) https://doi.org/10.1016/S0272-6963(98)00028-X
2	Operations managers must make decisions under uncertainty and incomplete information about future demand. The challenge is to develop reliable forecasts that account for trends,	Topic(s): Forecasting Discussion Question(s): ✓ The main objectives of forecasting. ✓ Discussion of different methods of forecasting?	Text Book: Heizer and Render, Chapter 4, Page: 108-117	Demand

	seasonality, and uncertainty to support effective capacity, resource, inventory, and operational planning.	✓ Qualitative methods of forecasting? ✓ Time-series and associative forecasting methods ✓ Exponential smoothening method of forecasting?	<i>How Companies Should Prepare Their Forecasts. Harvard Business Review Digital Article. Product #: H04WGS-PDF-ENG</i>	forecasting errors in industrial context: Measurement and impacts https://doi.org/10.1016/j.ijpe.2008.08.008
3	Operations managers need to minimize forecasting errors to improve demand estimates and decision accuracy. Selecting appropriate forecasting techniques that capture trends and seasonality is essential for reducing uncertainty and supporting reliable operational planning.	Topic(s): Forecasting Discussion Question(s): ✓ Discussion on Forecasting error. ✓ Different types of errors in forecasting? ✓ How do you select the best forecasting method? ✓ Role of seasonality trend in forecasting ✓ How exponential smoothening and trend combined to improve the forecasting accuracy?	Text Book: Heizer and Render, Chapter 4, Page: 118-135 Practice exercise with excel and solver	<i>Chinese Pharmaceuticals (HK) Limited: Effective Forecasting for Optimal Inventory Management. The Hong Kong University of Science and Technology. Product #: ST5-PDF-ENG</i>
4	The design function is constantly under pressure because to shifting consumer preferences and material advancements. As a result, the operations manager's job is crucial to ensuring a strong process for product development, meeting future expectations, and achieving cost competitiveness through techniques like value engineering of the house of quality.	Topic(s): Process and Product Design Discussion Question(s): ✓ Define a product development cycle. ✓ How to develop a house of quality? ✓ How did customers participate in the design of product and process? ✓ How to improve the value of a product or process? ✓ Value engineering and value analysis?	Text Book: Heizer and Render, Chapter 5, Page: 165-175	Value Engineering and Value Analysis https://www.youtube.com/watch?v=WLOKnW6bFU
5	To empower the practice of examining processes to find areas where they might be improved is a	Topic(s): Process Analysis Discussion Question(s): ✓ Important of process analysis	Text Book: Heizer and Render, Chapter 7,	How to Value

	very important decision for the operations managers. Operations managers need to have expertise in assessing their business processes and identify what is and isn't functioning inside their operations by using process analysis.	✓ Time function mapping ✓ Discussion on process charts. ✓ What is service blueprint ✓ What is value stream mapping.	Page: 298-312 <i>A Note on Process Analysis.</i> <i>Darden School of Business.</i> Product #: UV0425-PDF-ENG	Stream Map [STEP BY STEP] https://www.youtube.com/watch?v=7wD7R6x3Pv4
6	Organizations operating in global markets face complex decisions about where to locate facilities and operations. Managers must evaluate location strategies, market access, costs, resources, and infrastructure to select sites that strengthen operational efficiency and long-term competitiveness.	Topic(s): Location strategies Discussion Question(s): ✓ Important of location decision for organizations? ✓ Discussion on various factors that affect an organization's location decisions? ✓ Methods of evaluating the location alternatives? ✓ How do location decision strategies vary in the case of service operations? ✓ How to evaluate the effectiveness of location decisions?	Text Book: Heizer and Render, Chapter 8, Page: 340-353	How locations strategy helps companies with expansion and consolidation? https://lightcast.io/resources/blog/how-location-strategy-helps-companies-with-expansion-and-consolidation
7	Organizations must design layouts that optimize workflow, capacity, flexibility, cost, and productivity while supporting employee well-being and customer experience. Poor layout	Topic(s): Layout strategies Discussion Question(s): ✓ What makes layout choices crucial for businesses? ✓ Which are the various	Text Book: Heizer and Render, Chapter 9, Page: 370-391	<i>ABC Shipyard: The Facility Layout.</i> <i>Ivey Publishing.</i> Product #: W16803-PDF-ENG

	decisions can create long-term operational inefficiencies, higher costs, and an unfavorable work environment.	kinds of layouts? ✓ What distinguishing characteristics exist between process-oriented and fixed-position layouts? ✓ Which fundamental presumptions underlie repetitious and product-oriented layout?		Layout strategies https://www.youtube.com/watch?v=e2rzEcnm6Nw
8	Organizations face the challenge of completing projects within defined scope, time, cost, and quality constraints. Managers must effectively set objectives, allocate resources, manage risks, coordinate teams, and engage stakeholders to ensure successful project execution.	Topic(s): Importance of project management Discussion Question(s): ✓ Why project management is important for operations managers. ✓ Discussion on project life cycle. ✓ What is project scheduling ✓ Discussion on work breakdown structure. ✓ What is project controlling? ✓ What is cost overrun?	Text Book: Heizer and Render, Chapter 3, Page: 63-72	
9	Project managers must ensure projects remain on schedule, within budget, meet quality standards, and deliver stakeholder value. Effective management requires technical expertise in tools such as critical path analysis and probabilistic time estimation to manage uncertainty and complexity.	Topic(s): Project management techniques & Project Crashing Discussion Question(s): ✓ What are the techniques in project management? ✓ What is critical path? ✓ How to find out the critical path of the project? ✓ Discussion on activity on node and activity on arrow approaches. ✓ Discussion on forward pass and backward pass process to determine the critical path. ✓ What is project crashing? ✓ The importance of project crashing for the operations managers. ✓ Discussion on crash cost,	Text Book: Heizer and Render, Chapter 3, Page: 73-83 <i>Tastee Snax Cookie Company (A). Darden School of Business. Product #: UV5918-PDF-ENG</i>	<i>Tastee Snax Cookie Company (A). Darden School of Business. Product #: UV5918-PDF-ENG</i>

		normal cost, crash time, normal time. ✓ Discussion on calculation of crash cost per period. ✓ Consolidation of learnings		
10	Guest Lecture – Prof. Pramod Aggarwal			
